211

UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

UCSC

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2019/2020 – First Year Examination – Semester I – 2020

SCS1205 – Computer Systems

TWO (2) HOURS

Part A

To be completed by the candidate

Index No: ....

Important Instructions to candidates:

1. The medium of instruction and question is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the Question paper.
5. This paper has **04** questions and **15** pages.
6. Answer **ALL** questions within **two hours**. All questions carry equal marks (25 marks).
7. **This paper consists of two parts, Part A (Question No 1, Question No 2 and Question No 3, from Page 1 to 10) and Part B (Question No 4 from page 11 to 15) and submit separately.**
8. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
9. **Non-Programmable** calculators are **allowed**.

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Part A	
Question No	**Marks**
1	
2	
3	
Total	

Index No. : ....

Part A

Question 1

a. Identify the computer generation that corresponds with each of the following descriptions.

[4 Marks]

i. Mechanical calculating machines were invented and used.

ii. Started to use transistors in circuitry.

iii. Integrated circuits and silicon chips were developed to be used in circuitry.

iv. Computers used vacuum tubes and magnetic drums.

b. List three (3) key characteristics of *Von-Neumann architecture*?

[3 Marks]

c. Convert decimal number 101 into an **8-digit binary** representation. Show your calculation steps.

[2 Marks]

d. What is the decimal value of the **two's complement** binary number 1101011?

[3 Marks]

e. An IEEE 754 single precision value is written in hexa-decimal representation as 0xC1200000. What is the corresponding decimal value?

[5 Marks]

f. Convert IEEE 754 single precision number 0 10000101 01010100100000000000000 to decimal.

[5 Marks]

g. Explain ***rounding off error*** in floating point number representation using an appropriate example.

[3 Marks]

Index No. : ....

Question 2

a. Authorities of a university havw decided to measure body temperature of all students who enter to the university premises for examinations during this COVID-19 pandemic situation. Measurements are taken as follows.

- T1 – The first measurement of body temperature at the entrance.
- T2 – The second measurement is taken after 5 minutes **if only** T1 reports high temperature. A student who reports both T1 and T2 high, will not be allowed to enter the premises.
- T3 – The third measurement is taken during the examination.
- T4 – The fourth measurement is taken when a student is leaving the exam venue.

Based on the following rules, the medical officer will be alerted (X) on a student's temperature readings.

- T1 and T2 both report high temperatures.
- T3 reports high temperature while T1 or T2 has reported high.
- T4 reports high temperature while T3 has reported high.

i. By assuming high temperature of a measurement as Boolean value 1 and 0 otherwise, complete the following truth table for Boolean function X of *alerting the medical officer*. X is 1 when the *medical officer* is alerted and 0 otherwise.

[8 Marks]

T1	T2	T3	T4	X
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

ii. Identify all the cases where output is not defined (*don't care conditions*) in the above truth table and list all the don't care conditions in standard product form (*minterms*). State your reasons for the identification of don't care conditions.

[3 Marks]

iii. Derive the **most simplified** Boolean expression in **Standard Sum of Products form** for the above truth table using **Karnaugh Map**. Clearly show don't care conditions and your grouping.

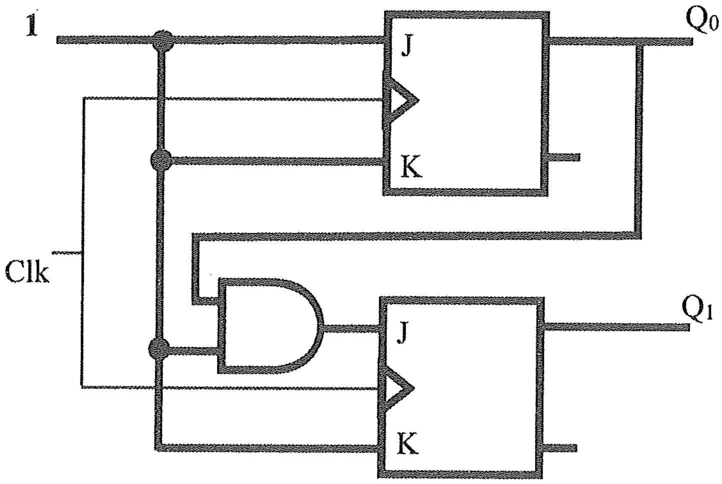
[5 Marks]

b. Using a suitable diagram, show how can a *Half Adder* be employed to construct *Half-Subtractor*?

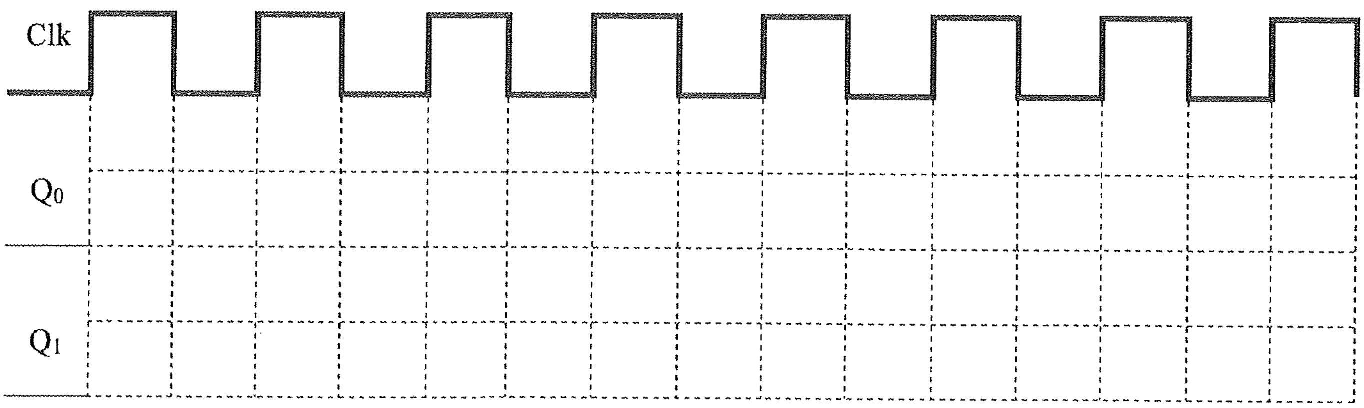
[4 Marks]

c. Consider the sequential logic circuit, which consists of two J-K flip flops and an AND gate.

[5 Marks]



Complete the timing diagram given below for the above sequential logic circuit. Assume both $\mathbf{Q_0}$ and $\mathbf{Q_1}$ are logic value zero (0) at the beginning.



Question 3

a. A computer has the following configurations.

- Address length is 32 bits
- A word size 8 bits
- 2MB cache volume
- Cache line size is 32KB

Answer the following questions by showing the calculation steps when necessary.

i. What is the maximum size of the memory this computer can have?

[2 Marks]

ii. How many words are in a single cache line?

[2 Marks]

iii. What is the maximum number of cache lines that can be held **simultaneously** at a given time?

[2 Marks]

iv. If the cache is *Direct Mapped*, what are the **bit lengths** of *Tag*, *Index* and *Offset* components of an address?

[4 Marks]

v. If the cache is ***4-Way Set Associative***, how many sets will be configured within the cache volume?

[2 Marks]

vi. If the cache is ***4-Way set Associative***, what are the **bit lengths** of Tag, Set Index and Offset components of an address?

[4 Marks]

vii. If the cache is ***2-Way Set Associative*** what are the bit lengths of Tag, Set Index and Offset components of an address?

[3 Marks]

b. Assume a *4-Way Set Associative* cache that employs **First In First Out** (FIFO) cache replacement policy. The CPU refers cache for four (4) different addresses A, B, C, and D **consecutively**, and each cache reference results a cache miss where it brings respective block of data (cache block) from the main memory to the cache.

i. Will we see a replacement of an already referred address by address references B, C, or D? Give reasons for your answer.

[3 Marks]

ii. Assume that the CPU now refers the address **E** which results another cache miss. If addresses A, B, C, D, and E map to the same set in the cache, which address of those five addresses should be referred next by the CPU to result a cache miss again? Justify your answer.

[3 Marks]

207

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SCS 1205 – Computer Systems – Part B

TWO (2) HOURS

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Examination Index No: ....

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1	
2	
3	
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Total	

Index No: ....

Part B

Question 4

(a) Suppose you are given two machines named A and B. If clock cycles of A is higher than clock cycles of B, can you claim that machine A is faster than B? Under what conditions clock speed can be used to compare the performance of two processors?

[5 Marks]

(b) Consider the following infix expression,

i. Convert the following infix expression into postfix notation. [3 Marks]

ii. Show calculation steps for evaluating the expression using stack instruction set architecture (ISA). [4 Marks]

$$\{(a.b) - (p + q)\}/(r - s)$$

Index No: ....

(c) Briefly explain the following components of a computer. Clearly state the functionality of each component.

(i) Control Unit

[2 Marks]

(ii) Memory Data Register (MDR)

[2 Marks]

(iii) Program Counter (PC)

[2 Marks]

(d) Suppose there exist a machine M where instructions are 16 bit long and word size of the machine is 8 bits and byte addressable. Instructions are in the form of <opcode> <register> <memory address>. Here, first 4 bits are allocated for opcode, next 4 bits are for the register and the remaining 16 bits are for the operand's memory address. Following are some listed opcodes for different types of instructions in M.

L R, A	**LOAD** the register **R** with the content of memory cell **A**
LI R,I	**LOAD** the register **R** with the value **I** $_{hex}$
LR R1,R2	**LOAD** the register **R1** with the content of register **R2**
ST R,A	**STORE** the content of register **R** to the memory cell whose address is A
ADD R0,R1,R2	**ADD** the numbers in registers **R1** and **R2** and place the result in **R0**
OR R0,R1,R2	**OR** the bit patterns in **R1** and **R2** and place the result in **R0**
XOR R0,R1,R2	**XOR** the bit patterns in **R1** and **R2** and place the result in **R0**
AND R0,R1,R2	**AND** the bit patterns in **R1** and **R2** and place the result in **R0**
JMP R,A	**JUMP** to the instruction located in the memory cell **A** if the bit pattern in **R** is equal to **R0.**
HALT	**HALT** the execution

Fill in the **missing** assembler code instructions to obtain the logic sequence given in the pseudocode below.

Initialize A to 20
Initialize B to 15
C is equal to the subtraction of B from A
If C is equal to 4
    then halt
else
    B is equal to B increment by 1

<u>Note :</u> Value of R0 register is 0. Assume that the values of A and B are given in decimal format in the pseudocode. Initial program counter (PC) is 30$_{hex}$.

[7 Marks]

....	LI 1, ...
....	LI 2, F
34	LI 0, ...
....	LI 3, FF
....	.... 4, ..., 3
40	LI 3, ...
....	ADD 2, ..., ...
....	.... 5, ..., ...
....	JMP 5,52
....	.... 2, ..., 3
....	JMP 2,38
52	HALT

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